MUHAMMAD EL-SAYEH, Ph.D.

Director of Secure AI & Infrastructure Systems · AI + Security Convergence Architect

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EXECUTIVE PROFILE

Multidisciplinary Ph.D. engineer and strategic leader with 15+ years architecting secure, intelligent systems across four convergent domains: AI/Agentic Systems, Cybersecurity (OT/ICS/Enterprise/Web3), Distributed Infrastructure, and Industrial Digital Transformation. Demonstrated at GlaxoSmithKline and Forbole Limited delivering institutional-grade outcomes - zero-finding regulatory audits, 99.99% infrastructure uptime, 80%+ SOC efficiency gains through AI automation - in mission-critical, compliance-intensive environments. Uniquely positioned at the intersection of AI systems engineering, enterprise security governance, blockchain infrastructure, and regulated industrial operations. Ph.D., Systems & Biomedical Engineering (Cairo University); Assistant Professor, Nile University.

CORE DOMAINS & COMPETENCIES

PILLAR 1 - AI Systems & Agentic Orchestration

‣ LLM Agent Orchestration (LangChain / LangGraph)	‣ MLOps Platforms (MLflow, ZenML, SageMaker)	‣ NLP & Clinical AI (cTAKES, DeepSpeech, Transformers)
‣ Computer Vision (YOLOv5/v8, DeepMedic, NiftyNet)	‣ Predictive Maintenance (LSTM, ARIMA, Autoencoders)	‣ GenAI Inference (Ollama, llama.cpp, vLLM, RAG)
‣ Feature Engineering & Model Lifecycle Management	‣ Distributed ML & HPC (MPI, SLURM, Spark)	‣ Vector DBs (FAISS, Chroma) & Multimodal Pipelines

PILLAR 2 - Cybersecurity (Enterprise · OT/ICS · Web3)

‣ ISMS Design & ISO 27001 Governance	‣ SOC Leadership & SIEM/SOAR Operations (ELK, Wazuh, Security Onion)	‣ HSM, PKI & MPC Key Management
‣ PAM & IAM (Vault, Keycloak, AWS IAM)	‣ IEC 62443 / ISA99 OT/ICS Cybersecurity	‣ Zero Trust Architecture & Network Segmentation
‣ Vulnerability Management & Penetration Testing	‣ Smart Contract Security Auditing (Foundry, Slither)	‣ Incident Response, DFIR & Threat Hunting

PILLAR 3 - Infrastructure & Platform Engineering

‣ Proxmox VE, Harvester HCI, Rancher / Kubernetes	‣ Multi-Region Cloud (AWS · GCP · Azure)	‣ Terraform, Ansible, Pulumi (IaC)
‣ CI/CD (GitHub Actions, GitLab CI, DevSecOps)	‣ Prometheus, Grafana, ELK Observability	‣ OPNsense Firewall, OpenVPN, DDoS Protection
‣ Go, Rust, Python API Layers (gRPC, WebSocket)	‣ Docker, Helm, Kubernetes Fleet Management	‣ Bare Metal Server Lifecycle (20+ servers, 350+ VMs)

PILLAR 4 - Industrial Digital Transformation & Compliance

‣ MES/OT Integration (ISA-95, OPC-UA, Modbus)	‣ SCADA/PLC (Siemens S7, Wonderware, DeltaV)	‣ GxP Compliance (21 CFR Part 11, GAMP 5, EU Annex 11)
‣ Computer System Validation (CSV) Lifecycle	‣ Digital Twin Design & Predictive Analytics	‣ Reliability Engineering (Weibull, FMECA, RCM)
‣ ISO 27001, NIST CSF, NIS2, DORA, PCI-DSS	‣ GDPR, Egyptian Data Protection Law, HIPAA	‣ Power BI, RPA, Mixed Reality Operational Tools

PROFESSIONAL EXPERIENCE

Head of Infrastructure & Cybersecurity · Forbole Limited *Dec 2021 – Present*

Blockchain Validator Company · Remote (Hong Kong / UK)

AI & Agentic Systems

- Designed and deployed Infratron - a centralized LLM agent orchestration platform (LangChain, LangGraph, Ollama, vLLM) managing 176 distributed validator nodes; implemented autonomous anomaly triage, runbook execution, alert routing, and human-in-the-loop guardrails.
- Reduced daily alert volume from 400+ signals to <15 prioritized incidents - 80%+ improvement in MTTR - through AI-driven correlation and automated SOAR-style response workflows.
- Built production NLP pipelines for automated smart contract vulnerability detection; trained transformer-based models for intelligent security event correlation across distributed infrastructure.
- Architected end-to-end MLOps pipelines (MLflow, ZenML) covering automated training, validation, versioning, and CI/CD deployment; implemented IaC compliance drift detection with auto-remediation PRs in GitHub.

Security Governance & Infrastructure

- Designed and enforced a comprehensive ISMS aligned with ISO 27001 and NIST CSF; led organizational risk assessments, gap analyses, and remediation roadmap ownership across IT, cloud, and blockchain layers.
- Designed and operate multi-region validator infrastructure across 170+ networks (Ethereum, Cosmos, Polkadot, Arbitrum, Optimism, Polygon): 99.99% uptime, zero slashing events across 176 active nodes.
- Provisioned and managed 20+ Proxmox VE bare metal servers hosting 350+ VMs; deployed Harvester HCI clusters integrated with Rancher for centralized Kubernetes fleet management.
- Configured and manage OPNsense firewalls, 4 OpenVPN tunnels, DDoS protection layers, network segmentation, and zero-trust access controls across AWS, GCP, and Azure.
- Implemented HSM-based (YubiHSM, AWS CloudHSM) and MPC institutional key management with threshold signatures, RBAC, and secure key rotation - achieving zero unauthorized access events.
- Constructed Go/Rust RPC/gRPC/WebSocket API layers via APISIX gateway; SAST/DAST integrated into CI/CD with mandatory security gates.
- Conducted 20+ smart contract security audits for DeFi/NFT protocols; integrated Foundry and Slither into GitHub Actions CI/CD for continuous automated contract testing.
- Conducted exploratory research into agentic offensive security frameworks (CAI, NeuroSploit) to validate defensive coverage and detection gap modeling.

OT/ICS Cybersecurity Champion & Digital Data Analytics Lead · GlaxoSmithKline (GSK) *Jan 2017 – Dec 2021*

Global Pharmaceutical Manufacturer · Giza, Egypt

Automation & Instrumentation Manager *Dec 2019 – Dec 2021*

Reliability & Automation/Instrumentation Section Head *Aug 2015 – Dec 2019*

Systems Validation & Calibration Section Head / Engineer *May 2011 – Aug 2015*

OT/ICS Security & Regulatory Compliance

- Led OT/ICS cybersecurity program for a major pharmaceutical manufacturing site; achieved 100% audit pass rate with zero critical or major findings across three regional regulatory audits (GMP, safety, corporate).
- Delivered cybersecurity program aligned with IEC 62443/ISA99, NIST CSF, ISO 27001, GDPR, HIPAA, and 21 CFR Part 11 covering 50+ control systems; governed data integrity under GAMP 5 and EU Annex 11 across 100+ systems.
- Designed defense-in-depth architecture: industrial DMZ, network segmentation, geo-zoning, SCADA hardening, and firewall policy migration aligned to IEC 62443 zoning.
- Developed portable AV scanning kiosks for air-gapped OT environments; maintained zero overdue security alerts throughout entire tenure.

AI / ML Production Deployments

- Engineered LSTM-based predictive maintenance models on vibration, temperature, and pressure sensor data; integrated with Infor EAM CMMS via REST APIs for automated work order generation - 25% reduction in unplanned equipment downtime.
- Deployed YOLOv5 computer vision quality inspection pipeline achieving 98% accuracy in GMP-regulated production, integrated with Wonderware SCADA for real-time line control.
- Deployed medical transcription NLP system (Apache cTAKES, DeepSpeech with custom medical vocabulary, 92% WER accuracy) and conversational AI (Mycroft AI) across production floor operations.
- Built NLP-powered RPA bots for automated batch tracking and deviation analysis - 40% reduction in manual reporting time.
- Led cross-functional Digital & Data Analytics programme; coached AI/data science champions across all departments, establishing ML best practices and Power BI, Power Apps, and RPA capabilities site-wide.

Automation & Instrumentation Management (Dec 2019 – Dec 2021)

- Managed cross-functional teams of 37 (4 specialists, 6 engineers, 27 technicians) across Maintenance Planning, Spare Parts, and Automation/Instrumentation departments for two manufacturing plants.
- Led computer system validation (CSV) framework covering 100+ GMP production and laboratory systems; achieved 100% remediation of data integrity gaps in compliance with 21 CFR Part 11, EU Annex 11, and GAMP 5.
- Directed data integrity assessments across all computerized systems: identified gaps, authored mitigation plans, coordinated control system upgrades, and led cross-functional assessment panels; served as CSQMS SME on multiple site committees.
- Drove instrumentation calibration programme, reducing overdue below 0.28% for external activities and to zero for in-house; reduced calibration audit findings by 85% against baseline.
- Governed CAPA lifecycle for all V&V-related quality events and deviations; ensured evidence-based closure with full traceability across all GMP systems.
- Specified and governed requirements for all computerized and control systems per defined global standards; led control system upgrade projects end-to-end with integrated security testing at every qualification stage.
- Passed GMP regional audit and Safety remediation audit with no significant, critical, or major findings; managed all engineering involvement in site regulatory inspections.
- Deployed mixed reality training apps, Power BI OEE dashboards, RPA for batch tracking automation, and mobile apps for safety reporting as part of enterprise Digital Transformation roadmap.

Reliability & Automation/Instrumentation Section Head (Aug 2015 – Dec 2019)

- Introduced computerized maintenance management system (Infor EAM CMMS) across 1,200+ production and utility assets; implemented risk-based maintenance strategy using Weibull distribution, moment estimation, and Criticality Assessment - 50% reduction in instrumentation failure rate and HVAC breakdowns.
- Designed and deployed Value-Driven Maintenance (VDM) continuous improvement framework; launched condition-based maintenance (CBM) pilots and RCM-aligned maintenance strategies across both manufacturing plants.
- Developed functional test procedures and alarm management optimization for critical control modules; contributed to data integrity gap analysis and control system upgrade planning.
- Built Power BI KPI dashboards monitoring OEE, schedule adherence, breakdowns, MTBF, calibration compliance, and performance trends - direct precursor to Digital Twin analytics and real-time security telemetry design.

- Managed Siemens S7-300/400/1200/1500 PLC systems (TIA Portal, WinCC), Wonderware SCADA, SIMATIC Batch/Route Control DCS, and calibration programs across two plants.
- Led robotics and automation projects; managed change management and community-of-practice across multiple sites; drove cross-departmental alignment across Engineering, Quality, Validation, and Production.

Systems Validation & Calibration Section Head / Engineer (May 2011 – Aug 2015)

- Established computer system validation (CSV) lifecycle methodology and software testing approach for legacy system revalidation and new system commissioning; managed and approved Site Validation Master Plan.
- Directed a team of 2 senior engineers, 6 junior engineers, and 2 technicians across all qualification and calibration activities; provided technical mentorship and performance management throughout.
- Led all sterility assurance qualification activities for sterilizers, clean rooms, and depyrogenation tunnels per ISO 11138, ISO 17665, EN 285, and ISO 14644; delivered fully documented IQ/OQ/PQ packages with zero critical findings.
- Introduced software testing methodologies (functional, integration, regression, and security) into validation and qualification test cases - a practice that predates and directly informed the team's later AI/ML validation and cybersecurity framework.
- Reduced production environment deviations by 10% annually through streamlined validation processes; launched paperless calibration systems reducing overall project lifecycle by 14 days.
- Improved cross-departmental alignment across Engineering, Quality, Validation, and Production for introduction of new computerized systems; authored SOPs, protocols, summary reports, and CAPAs throughout.

R&D Engineer - RFID/RTLS & Embedded Systems · Crescent Technologies *Aug 2009 – May 2011*
Giza, Egypt

- Designed novel algorithm for RFID-based tagless indoor object localization using COTS tags; patented and published via IEEE.
- Developed anti-counterfeit and anti-infant abduction systems using RF-SAW circuits; built OCR system for blood transfusion bag UDI detection with full functional, integration, load, regression, and security testing.

ACADEMIC APPOINTMENTS

Assistant Professor - Biomedical Informatics · Nile University *Oct 2023 – Present*
Giza, Egypt

- Teaching: Advanced Python for Bioinformatics, Biomedical Data Analysis & Visualization, Data Mining and Machine Learning.
- Supervising postgraduate research in medical image analysis, clinical NLP, AI-driven diagnostics, LLM applications, and clinical decision support.

Assistant Professor - Biomedical Engineering · Misr University for Science and Technology (MUST) *Jan 2017 – Jul 2023*
Cairo, Egypt

- Taught Nuclear Physics, Radiological Medical Equipment, nano-biosensors, and nanoelectronics; supervised 15+ graduation projects in ML-powered medical devices, robotics, and embedded biomedical systems.

Adjunct Assistant Professor - Pharmaceutical Engineering · German International University (GIU) *Aug 2022 – Jul 2023*
Cairo, Egypt

- Taught Quality Management Systems, Quality Risk Management, Pharmaceutical Process Validation, and Biologics Manufacturing.

SELECTED PROJECTS & RESEARCH

Infratron - AI-Orchestrated Infrastructure Operations Platform

- Centralized LLM reasoning platform (LangChain, LangGraph, Ollama) managing 176 distributed nodes with tool calling, state management, memory, autonomous triage, runbook execution, and human-in-the-loop guardrails.
- IaC compliance drift detection: continuous comparison of live infrastructure against Terraform baselines; auto-raises remediation PRs in GitHub on policy deviation.

Kastle Wallet - Full Security Audit & Vulnerability Disclosure Programme

- Conducted SAST (OWASP Top 10) on browser extension and DAST (Frida-based) on Android wallet; identified multiple critical-severity findings; delivered phased remediation roadmap reducing open criticals from 6 to 0 within 30 days.
- Established end-to-end VDP: severity tiers, responsible disclosure policy, PGP-encrypted reporting channel, SLAs, and safe-harbor language.

ARC-AGI Solver - Hybrid Symbolic/Neural Program Synthesis

- Developed hybrid DSL compositional search (depth=5, beam=500) combined with LLM-based reasoning; applied structured tool use and program synthesis for complex multi-step abstract reasoning at frontier research level.
- Currently extending with V-JEPA + GNCA + LLM Planner architecture for visual-spatial reasoning benchmarks.

Medical Image Analysis & Clinical AI

- Total brain parcellation and volumetric analysis using NiftyNet (HighRes3DNet); semantic segmentation of MS lesions using DeepMedic CNN achieving 95% Dice score.
- Medical transcription system combining DeepSpeech with custom medical vocabulary achieving 92% WER accuracy; deployed in live GMP pharmaceutical production environment.

AEGIS - EEG-Based Neural Authentication System (Research)

- Motor imagery EEG cryptographic authentication system: neural signal acquisition, feature extraction pipeline, and threshold cryptographic signing - demonstrating deep BCI/neural AI pipeline expertise.

Single-Cell Nuclear Phenotyping - Ongoing Research

- Supervised ML pipeline (XGBoost with multi-scale chromatin feature extraction) on DAPI-derived nuclear morphology data from AC16 cardiomyocyte datasets (IDR Project 1304); full mathematical feature engineering formalism for nuclear phenotype classification.

Mini-HPC Cluster - Distributed ML on Genomic Data

- 3-node HPC cluster (SLURM, MPI, mpi4py) on Ubuntu 22.04; distributed ML pipeline for RNA-seq gene expression classification (leukemia vs. normal, GSE65133) using PCA, per-node Random Forest, and majority-vote ensemble aggregation.

Biomedical AI - Drug Discovery & Bioinformatics

- In-silico antiviral agent design for SARS-CoV-2 via molecular docking and ML; protein structure prediction using transformer-based models; microbiome metabolic modeling framework (GMrepo, AGORA, ODE-based IL-10/NF-κB pathway modeling).

TECHNICAL SKILLS

AI / GenAI	LangChain, LangGraph, Ollama, llama.cpp, vLLM, OpenAI API, Hugging Face, RAG, FAISS, Chroma, V-JEPA, GNCA
ML Frameworks	TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, LightGBM, MLflow, ZenML, W&B, Spark ML, Nextflow

NLP	Transformers (BERT, GPT, T5), spaCy, NLTK, Apache cTAKES, DeepSpeech, NER, RAG, clinical NLP
Computer Vision	YOLOv5/v8, CNNs, ViT, DeepMedic, NiftyNet, OpenCV, medical image segmentation & parcellation
Security Tools	ELK/SIEM, Wazuh, Security Onion, TheHive, MISP, Shuffle SOAR, Suricata, Zeek, Velociraptor, Wireshark
Security Infra	YubiHSM, AWS CloudHSM, MPC/Threshold Sigs, PKI, Keycloak, HashiCorp Vault, AWS IAM, Azure AD, APISIX WAF
Vuln / AppSec	Nmap, OpenVAS, Metasploit, Burp Suite, OWASP ZAP, Frida, Snyk, SonarQube, Slither, Foundry, Mythril, Echidna
Infrastructure	Proxmox VE, Harvester HCI, Rancher, Kubernetes, Docker, Helm; OPNsense, OpenVPN, pfSense; NFS, Samba
Cloud	AWS (SageMaker, GuardDuty, CloudHSM, Security Hub, Inspector), Azure ML + Security Center, GCP AI Platform
IaC / DevSecOps	Terraform, Ansible, GitHub Actions, GitLab CI, Vault secrets management, shift-left security gates
Observability	Prometheus, Grafana, ELK Stack, Security Onion, AI-driven alert correlation; Prometheus exporters at scale
Industrial / OT	Siemens S7 (TIA Portal, WinCC), Rockwell Allen-Bradley, Wonderware, DeltaV, Yokogawa; OPC-UA, Modbus, PROFINET, DNP3, MQTT, BACnet, HART
Programming	Python (Expert), Go, Rust, C/C++, Bash, Solidity, JavaScript/TypeScript, MATLAB, R, Java, SQL/NoSQL
Data Engineering	Apache Spark, PySpark, Hadoop, Dask; AWS HealthOmics, HealthLake; SageMaker Feature Store; Athena
Blockchain	Cosmos SDK, Tendermint, IBC, Ethereum/EVM, CosmWasm, Solidity, ZK-SNARKs, MPC, Multi-sig

STANDARDS & COMPLIANCE EXPERTISE

InfoSec	ISO/IEC 27001, NIST CSF, NIS2, DORA, ENISA, ISO/IEC 27005, NIST RMF
OT / ICS	IEC 62443 / ISA99 (full series), DHS ICS-CERT certified, SCADA/DCS zoning & hardening
GxP / GMP	21 CFR Part 11, GAMP 5, EU Annex 11, ISA-95, ISO 13485, ISO 14644, ISO 11138, EN 285
Privacy	GDPR, Egyptian Data Protection Law, HIPAA, PCI-DSS
Medical Device	ISO 14971 (Risk Mgmt), IEC 62304 (Software Lifecycle), Medical Devices Regulation & Technical File

EDUCATION

Ph.D. - Systems and Biomedical Engineering *Grade: Excellent*
Cairo University, Egypt *2011 – 2016*
Dissertation: Qualitative and Quantitative Detection of Bacteria and Bacterial Endotoxin Using Raman and Microwave Spectroscopy

M.Sc. - Systems and Biomedical Engineering *Grade: Excellent*
Cairo University, Egypt · Thesis: Real-Time Locating Systems and RFID – Applications in Healthcare Sectors *2008 – 2011*

B.Sc. - Biomedical Engineering (GPA: 3.74)
Higher Technological Institute, 10th of Ramadan City, Egypt *2003 – 2008*

CERTIFICATIONS & TRAINING

‣ EC-Council Security Analyst (ESCA) - Certified	‣ EC-Council Advanced Penetration Testing (LPT) - Trained	‣ CISSP Examination Preparatory Course (8 Domains)
‣ DHS ICS-CERT: ICS Cybersecurity Full Series (100W · 210W-01–10 · 300)	‣ OPSEC for Control Systems - DHS ICS-CERT	‣ Certified Data Scientist Professional (CDSP) - EPSILONAI, USA
‣ Deep Learning Specialization - Coursera / Stanford	‣ MLOps for Production - Coursera / DeepLearning.AI	‣ ISO/IEC 27001 ISMS Design & Audit Preparation
‣ Smart Contract Security & Auditing (Practical)	‣ Medical Devices Regulation & Technical File Preparation	‣ GAMP 5 & GxP Compliance (Validated Systems)

SELECTED PUBLICATIONS & PATENTS

- "Cybersecurity Architecture for the Internet of Medical Things and Connected Devices Using Blockchain," Biomedical Engineering: Applications, Basis, and Communications, Vol. 33, No. 02, March 2021.
- "Detection and Identification System of Bacteria and Bacterial Endotoxin Based on Raman Spectroscopy," International Journal of Advanced Computer Science and Applications (IJACSA), 7(3), April 2016.
- "Ultra-Wide Band Based Quantitative and Qualitative Method for Bacterial Endotoxin Detection," Journal of Microwave Power and Electromagnetic Energy, 49(1), February 2015.
- RFID-Based Tag-less Object Localization Using Back Projection - Patent & IEEE Publication.

PROFESSIONAL MEMBERSHIPS & AWARDS

Memberships: International Society for Automation (ISA) - Active Member · International Society of Pharmaceutical Engineering (ISPE) - Active Member · Medical Devices Technical & Regulatory Consultant · Aseptic Engineering Subject Matter Expert

Awards: GSK Innovation-Performance-Trust (IPT) Award, 2019 · GSK R&D Bronze Award (Top Idea, IGNITE Program), 2013 · Robocon Egypt, 2007

Languages: Arabic (Native) · English (Proficient - all skills) · French (Basic)

All claims herein are factual and verifiable. References available on request.